

# **C/C++ SDK User Manual**

**V1.0**

## 目录

|                                                  |    |
|--------------------------------------------------|----|
| 1. Introduction .....                            | 4  |
| 2. System Requirements .....                     | 4  |
| 3. Installation and Uninstallation .....         | 4  |
| 4. Quick Start .....                             | 4  |
| 4.1. SDK Development Workflow .....              | 4  |
| 4.2. Simple Usage Example .....                  | 5  |
| 5. Interface Definitions .....                   | 7  |
| 5.1. Callback Definitions .....                  | 7  |
| 5.2. Function Reference .....                    | 8  |
| 5.2.1. Create device instance. ....              | 8  |
| 5.2.2. Destroy Device Instance .....             | 9  |
| 5.2.3. Open Device .....                         | 9  |
| 5.2.4. Close Device .....                        | 10 |
| 5.2.5. Check if Device Connection is Open .....  | 10 |
| 5.2.6. Check if Device Connection is Valid ..... | 10 |
| 5.2.7. Restart the RFID reader. ....             | 11 |
| 5.2.8. Reset Device .....                        | 11 |
| 5.2.9. Query Device Parameters .....             | 11 |
| 5.2.10. Query RFID Parameters .....              | 12 |
| 5.2.11. Query UHF Parameters .....               | 12 |
| 5.2.12. Query Application Parameters .....       | 13 |
| 5.2.13. Set Device Parameters .....              | 13 |
| 5.2.14. Set RFID Parameters .....                | 14 |
| 5.2.15. Set UHF Parameters .....                 | 14 |
| 5.2.16. Set Application Parameters .....         | 15 |
| 5.2.17. Query GPIO Mode .....                    | 15 |
| 5.2.18. Set GPIO Mode .....                      | 16 |
| 5.2.19. Query GPIO Logic Level State .....       | 16 |
| 5.2.20. Set GPIO Logic Level State .....         | 17 |
| 5.2.21. Buzzer Beep .....                        | 17 |
| 5.2.22. Start Asynchronous RFID Inventory .....  | 17 |
| 5.2.23. Stop RFID .....                          | 18 |
| 5.2.24. Read RFID Tag Data .....                 | 18 |
| 5.2.25. Write RFID Tag Data .....                | 19 |

|                                       |    |
|---------------------------------------|----|
| 5.2.26. Lock RFID Tag .....           | 19 |
| 5.2.27. Kill RFID Tag .....           | 20 |
| 6. Type Definitions .....             | 20 |
| 6.1. Enumerations .....               | 20 |
| 6.1.1. device_gpio_index_t .....      | 20 |
| 6.1.2. device_gpio_mode_t .....       | 22 |
| 6.1.3. device_status_t .....          | 22 |
| 6.1.4. antenna_index_t .....          | 24 |
| 6.1.5. antenna_mask_t .....           | 24 |
| 6.1.6. uhf_bank_type_t .....          | 25 |
| 6.1.7. uhf_lock_bank_type_t .....     | 25 |
| 6.1.8. rfid_lock_type_t .....         | 26 |
| 6.2. Structures .....                 | 26 |
| 6.2.1. rfid_inventory_t .....         | 27 |
| 6.2.2. rfid_kill_t .....              | 28 |
| 6.2.3. rfid_lock_t .....              | 29 |
| 6.2.4. rfid_read_t .....              | 29 |
| 6.2.5. rfid_write_t .....             | 30 |
| 6.2.6. device_rfid_tag_t .....        | 30 |
| Appendix 1: Parameter Table .....     | 31 |
| 1.1 Device Parameters .....           | 31 |
| 1.2 RFID Parameters .....             | 36 |
| 1.3 UHF Parameters .....              | 40 |
| 1.4 Application Parameters .....      | 42 |
| Appendix 2: Region Table .....        | 45 |
| Appendix 3: Device Status Codes ..... | 46 |

# 1. Introduction

The SDK is a set of APIs specifically designed for our company's devices, intended to help users efficiently and rapidly develop device applications.

## 2. System Requirements

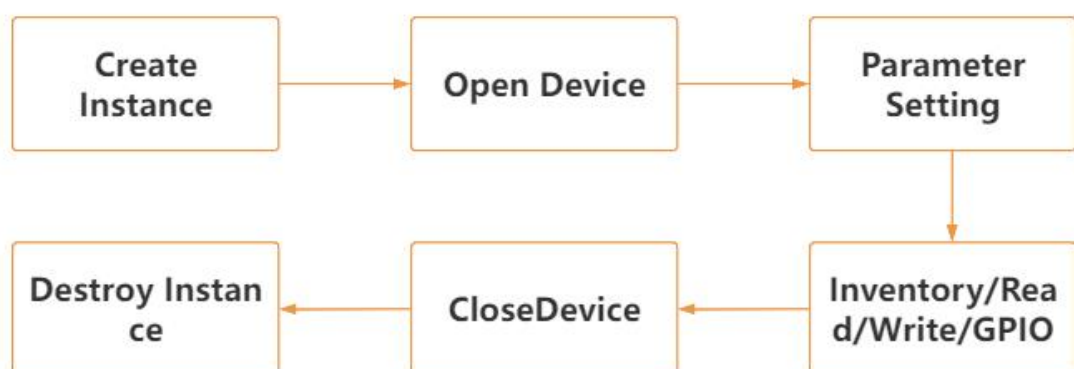
- ✓ Operating System: Windows/Linux

## 3. Installation and Uninstallation

The SDK is provided as a library and requires no installation.

## 4. Quick Start

### 4.1. SDK Development Workflow



## 4.2. Simple Usage Example

```
/**
 * @file example_base_rfid_inventory.c
 * @brief C Example: Basic RFID Inventory
 *
 * Compile:
 * gcc -o example_base_inventory example_base_inventory.c -ldevice_api
 */

#include "device_api.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

/* RFID Tag Callback */
void on_rfid_tag(device_handle_t dev, const device_rfid_tag_t* tag, void* userdata)
{
    (void)dev;
    (void)userdata;

    char epc_hex[128];
    for (int i = 0; i < tag->epc_len; i++) {
        sprintf(epc_hex + i * 2, "%02X", tag->epc[i]);
    }
    epc_hex[tag->epc_len * 2] = '\0';

    printf("Tag: EPC=%s, RSSI=%.2f dBm, Ant=%d\n",
        epc_hex, tag->rssi * 0.01f, (int)tag->antenna);
}

void on_rfid_stop(device_handle_t dev, uint8_t reason, void* userdata) {
    (void)dev;
    (void)userdata;
    printf("RFID inventory stopped. Reason: 0x%02X\n", reason);
}

/* Disconnect callback */
void on_disconnect(device_handle_t dev, int will_reconnect, void* userdata) {
    (void)dev;
    (void)userdata;
    printf("Disconnected! Will reconnect: %s\n", will_reconnect ? "yes" : "no");
}

/* Reconnect callback */
void on_reconnected(device_handle_t dev, void* userdata) {
    (void)dev;
    (void)userdata;
    printf("Reconnected!\n");
}

/* Log callback */
void on_log(device_handle_t dev, const char* msg, void* userdata) {
    (void)dev;
    (void)userdata;
}
```

```

    printf("[LOG] %s", msg);
}

int main(int argc, char* argv[]) {
    const char* port = (argc > 1) ? argv[1] : "192.168.1.201";

    /* Set callbacks */
    device_callbacks_t cbs;
    memset(&cbs, 0, sizeof(cbs));
    cbs.on_rfid_tag = on_rfid_tag;
    cbs.on_rfid_stop = on_rfid_stop;
    cbs.on_disconnect = on_disconnect;
    cbs.on_reconnected = on_reconnected;
    // cbs.on_log = on_log;

    /* Create device instance */
    device_handle_t dev = device_create(&cbs);
    if (!dev) {
        fprintf(stderr, "Failed to create device\n");
        return 1;
    }

    /* Open device */
    printf("Opening %s ...\n", port);
    device_status_t status = device_open(dev, port, 1);
    if (status != DEVICE_OK) {
        fprintf(stderr, "Failed to open device: 0x%02X\n", (int)status);
        device_destroy(dev);
        return 1;
    }
    printf("Device opened successfully!\n");

    /* Set RFID antenna */
    printf("Setting RFID antenna...\n");
    device_param_value_t antenna;
    param_reset(&antenna);
    param_append_uint(&antenna, ANTENNA_MASK_1|ANTENNA_MASK_4); // Working
    antenna mask
    param_append_ushort(&antenna, 0); // Antenna dwell time

    status = device_save_rfid_param(dev, RFID_ANTENNA, &antenna);
    if (status != DEVICE_OK) {
        fprintf(stderr, "Failed to set rfid antenna: 0x%02X\n", (int)status);
        device_destroy(dev);
        return 1;
    }
    printf("Antenna set successfully.\n");

    /* Start RFID inventory */
    printf("Starting RFID inventory...\n");
    status = device_rfid_inventory(dev, NULL);
    if (status != DEVICE_OK) {
        fprintf(stderr, "Failed to start inventory: 0x%02X\n", (int)status);
        device_destroy(dev);
        return 1;
    }
    printf("Inventory started.\n");

    /* Wait for user input to stop inventory */
    printf("Press Enter to stop...\n");
}

```

```

    getchar();

    /* Stop inventory */
    device_rfid_stop(dev);

    /* Close device connection */
    device_close(dev);
    device_destroy(dev);

    printf("Done.\n");
    return 0;
}

```

## 5. Interface Definitions

The device access interface covers operations such as opening and closing the device, querying/configuring parameters, GPIO control, device firmware upgrade, RFID inventory, and RFID tag read/write operations.

### 5.1. Callback Definitions

| Event          | Parameters                                                                                      | Description                        |
|----------------|-------------------------------------------------------------------------------------------------|------------------------------------|
| on_log         | dev: device handle<br>userdata: user data passed to callback<br>msg: log message                | SDK log callback                   |
| on_reconnected | dev: device handle<br>userdata: user data passed to callback                                    | Device reconnection callback (TCP) |
| on_disconnect  | dev: device handle<br>userdata: user data passed to callback<br>will_reconnect: whether the API | Device disconnection callback      |

| Event           | Parameters                                                                                                      | Description                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                 | will attempt reconnection                                                                                       |                                   |
| on_rfid_tag     | dev: device handle<br>userdata: user data passed to callback<br>tag: tag data                                   | RFID tag data received callback   |
| on_rfid_stop    | dev: device handle<br>userdata: user data passed to callback<br>reason: RFID stop reason                        | RFID inventory stopped callback   |
| on_gpio_event   | dev: device handle<br>userdata: user data passed to callback<br>gpio: GPIO index (1-based)<br>state: GPIO state | Device GPIO state change callback |
| on_ota_progress | dev: device handle<br>userdata: user data passed to callback<br>percent: OTA progress percentage                | Firmware OTA progress callback    |

Note: To improve SDK processing efficiency, do not perform long-running business logic in event handler functions. The application should only receive data and forward it to other threads for processing.

## 5.2. Function Reference

### 5.2.1. Create device instance.

|                                                                           | Type               | Description           |
|---------------------------------------------------------------------------|--------------------|-----------------------|
| <code>device_handle_t device_create(const device_callbacks_t* cbs)</code> |                    |                       |
| Parameter                                                                 |                    |                       |
| cbs                                                                       | device_callbacks_t | Callback function set |



|                         |                 |               |
|-------------------------|-----------------|---------------|
| Return Value            |                 |               |
|                         | device_handle_t | Device handle |
| Description             |                 |               |
| Create device instance. |                 |               |

## 5.2.2. Destroy Device Instance

|                                          | Type            | Description   |
|------------------------------------------|-----------------|---------------|
| void device_destroy(device_handle_t dev) |                 |               |
| Parameter                                |                 |               |
| dev                                      | device_handle_t | Device handle |
| Return Value                             |                 |               |
| None                                     |                 |               |
| Description                              |                 |               |
| Destroy device instance and free memory. |                 |               |

## 5.2.3. Open Device

|                                                                                   | Type            | Description                                                                                                                   |
|-----------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| device_status_t device_open(device_handle_t dev, const char* param, uint8_t addr) |                 |                                                                                                                               |
| Parameter                                                                         |                 |                                                                                                                               |
| dev                                                                               | device_handle_t | Device handle                                                                                                                 |
| param                                                                             | char*           | Connection parameter:<br>Serial: [port:baudrate]<br>TCP: [IP:port]<br>E.g.:<br>Serial: COM1:115200<br>TCP: 192.168.1.201:8090 |
| addr                                                                              | uint8_t         | Communication address, valid range 1~250, 255 is broadcast address                                                            |
| Return Value                                                                      |                 |                                                                                                                               |
|                                                                                   | device_status_t | Device open result                                                                                                            |

|                                                                  |
|------------------------------------------------------------------|
| Description                                                      |
| Open the device connection and perform necessary initialization. |

#### 5.2.4. Close Device

|                                        | Type            | Description   |
|----------------------------------------|-----------------|---------------|
| void device_close(device_handle_t dev) |                 |               |
| Parameter                              |                 |               |
| dev                                    | device_handle_t | Device handle |
| Return Value                           |                 |               |
| None                                   |                 |               |
| Description                            |                 |               |
| Disconnect the device.                 |                 |               |

#### 5.2.5. Check if Device Connection is Open

|                                                                                                  | Type            | Description                                   |
|--------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------|
| int device_is_connected(device_handle_t dev)                                                     |                 |                                               |
| Parameter                                                                                        |                 |                                               |
| dev                                                                                              | device_handle_t | Device handle                                 |
| Return Value                                                                                     |                 |                                               |
|                                                                                                  | int             | Returns whether the device connection is open |
| Description                                                                                      |                 |                                               |
| Only checks whether the current connection is open; does not perform communication verification. |                 |                                               |

#### 5.2.6. Check if Device Connection is Valid

|                                                           | Type            | Description                                |
|-----------------------------------------------------------|-----------------|--------------------------------------------|
| device_status_t device_check_connect(device_handle_t dev) |                 |                                            |
| Parameter                                                 |                 |                                            |
| dev                                                       | device_handle_t | Device handle                              |
| Return Value                                              |                 |                                            |
|                                                           | device_status_t | Returns the device connection check result |
| Description                                               |                 |                                            |

|                                                                         |
|-------------------------------------------------------------------------|
| Check whether the current connection is valid via a communication test. |
|-------------------------------------------------------------------------|

### 5.2.7. Restart the RFID reader.

|                                                     | Type            | Description                |
|-----------------------------------------------------|-----------------|----------------------------|
| device_status_t device_restart(device_handle_t dev) |                 |                            |
| Parameter                                           |                 |                            |
| dev                                                 | device_handle_t | Device handle              |
| Return Value                                        |                 |                            |
|                                                     | device_status_t | Returns the restart result |
| Description                                         |                 |                            |
| Restart the RFID reader.                            |                 |                            |

### 5.2.8. Reset Device

|                                                     | Type            | Description                     |
|-----------------------------------------------------|-----------------|---------------------------------|
| device_status_t device_restore(device_handle_t dev) |                 |                                 |
| Parameter                                           |                 |                                 |
| dev                                                 | device_handle_t | Device handle                   |
| Return Value                                        |                 |                                 |
|                                                     | device_status_t | Returns the device reset result |
| Description                                         |                 |                                 |
| Reset the device to factory defaults.               |                 |                                 |

### 5.2.9. Query Device Parameters

|                                                                                                             | Type                  | Description                                                 |
|-------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------|
| device_status_t device_get_device_param(device_handle_t dev, uint8_t id, device_param_value_t* param_value) |                       |                                                             |
| Parameter                                                                                                   |                       |                                                             |
| dev                                                                                                         | device_handle_t       | Device handle                                               |
| id                                                                                                          | uint8_t               | Device parameter type                                       |
| param_val                                                                                                   | device_param_value_t* | Parameter object used to store the queried parameter value. |
| Return Value                                                                                                |                       |                                                             |
|                                                                                                             | device_status_t       | Returns the query result                                    |

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| Description                                                                                           |
| Query device parameters. See the Device Parameters section for parameter definitions and value types. |

### 5.2.10. Query RFID Parameters

|                                                                                                           | Type                  | Description                                                 |
|-----------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------|
| device_status_t device_get_rfid_param(device_handle_t dev, uint8_t id, device_param_value_t* param_value) |                       |                                                             |
| Parameter                                                                                                 |                       |                                                             |
| dev                                                                                                       | device_handle_t       | Device handle                                               |
| id                                                                                                        | uint8_t               | RFID parameter type                                         |
| param_val                                                                                                 | device_param_value_t* | Parameter object used to store the queried parameter value. |
| Return Value                                                                                              |                       |                                                             |
|                                                                                                           | device_status_t       | Returns the query result                                    |
| Description                                                                                               |                       |                                                             |
| Query RFID parameters. See the RFID Parameters section for parameter definitions and value types.         |                       |                                                             |

### 5.2.11. Query UHF Parameters

|                                                                                                          | Type                  | Description                                                 |
|----------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------|
| device_status_t device_get_uhf_param(device_handle_t dev, uint8_t id, device_param_value_t* param_value) |                       |                                                             |
| Parameter                                                                                                |                       |                                                             |
| dev                                                                                                      | device_handle_t       | Device handle                                               |
| id                                                                                                       | uint8_t               | UHF parameter type                                          |
| param_val                                                                                                | device_param_value_t* | Parameter object used to store the queried parameter value. |
| Return Value                                                                                             |                       |                                                             |
|                                                                                                          | device_status_t       | Returns the query result                                    |
| Description                                                                                              |                       |                                                             |
| Query UHF parameters. See the UHF Parameters section for parameter definitions and value types.          |                       |                                                             |

## 5.2.12. Query Application Parameters

|                                                                                                                       | Type                  | Description                                                 |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------|
| <code>device_status_t device_get_app_param(device_handle_t dev, uint8_t id, device_param_value_t* param_value)</code> |                       |                                                             |
| Parameter                                                                                                             |                       |                                                             |
| dev                                                                                                                   | device_handle_t       | Device handle                                               |
| id                                                                                                                    | uint8_t               | Application parameter type                                  |
| param_val                                                                                                             | device_param_value_t* | Parameter object used to store the queried parameter value. |
| Return Value                                                                                                          |                       |                                                             |
|                                                                                                                       | device_status_t       | Returns the query result                                    |
| Description                                                                                                           |                       |                                                             |
| Query application parameters. See the Application Parameters section for parameter definitions and value types.       |                       |                                                             |

## 5.2.13. Set Device Parameters

|                                                                                                                                                                                                                                                                              | Type                  | Description                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------|
| <code>device_status_t device_set_device_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value, bool save)</code><br><code>device_status_t device_save_device_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value)</code> |                       |                                                               |
| Parameter                                                                                                                                                                                                                                                                    |                       |                                                               |
| dev                                                                                                                                                                                                                                                                          | device_handle_t       | Device handle                                                 |
| id                                                                                                                                                                                                                                                                           | uint8_t               | Device parameter type                                         |
| param_val                                                                                                                                                                                                                                                                    | device_param_value_t* | Parameter object used to store the parameter value to be set. |
| saved                                                                                                                                                                                                                                                                        | bool                  | Whether to persist (save to device Flash)                     |
| Return Value                                                                                                                                                                                                                                                                 |                       |                                                               |
|                                                                                                                                                                                                                                                                              | device_status_t       | Returns the set result                                        |
| Description                                                                                                                                                                                                                                                                  |                       |                                                               |
| Set device parameters. See the Device Parameters section for parameter definitions and value types.                                                                                                                                                                          |                       |                                                               |

## 5.2.14. Set RFID Parameters

|                                                                                                                                                                                                                                                                          | Type                  | Description                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------|
| <code>device_status_t device_set_rfid_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value, bool save)</code><br><code>device_status_t device_save_rfid_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value)</code> |                       |                                                               |
| Parameter                                                                                                                                                                                                                                                                |                       |                                                               |
| dev                                                                                                                                                                                                                                                                      | device_handle_t       | Device handle                                                 |
| id                                                                                                                                                                                                                                                                       | uint8_t               | RFID parameter type                                           |
| param_val                                                                                                                                                                                                                                                                | device_param_value_t* | Parameter object used to store the parameter value to be set. |
| saved                                                                                                                                                                                                                                                                    | bool                  | Whether to persist (save to device Flash)                     |
| Return Value                                                                                                                                                                                                                                                             |                       |                                                               |
|                                                                                                                                                                                                                                                                          | device_status_t       | Returns the set result                                        |
| Description                                                                                                                                                                                                                                                              |                       |                                                               |
| Set RFID parameters. See the RFID Parameters section for parameter definitions and value types.                                                                                                                                                                          |                       |                                                               |

## 5.2.15. Set UHF Parameters

|                                                                                                                                                                                                                                                                        | Type                  | Description                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------|
| <code>device_status_t device_set_uhf_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value, bool save)</code><br><code>device_status_t device_save_uhf_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value)</code> |                       |                                                               |
| Parameter                                                                                                                                                                                                                                                              |                       |                                                               |
| dev                                                                                                                                                                                                                                                                    | device_handle_t       | Device handle                                                 |
| id                                                                                                                                                                                                                                                                     | uint8_t               | UHF parameter type                                            |
| param_val                                                                                                                                                                                                                                                              | device_param_value_t* | Parameter object used to store the parameter value to be set. |
| saved                                                                                                                                                                                                                                                                  | bool                  | Whether to persist (save to device Flash)                     |
| Return Value                                                                                                                                                                                                                                                           |                       |                                                               |
|                                                                                                                                                                                                                                                                        | device_status_t       | Returns the set result                                        |
| Description                                                                                                                                                                                                                                                            |                       |                                                               |

Set UHF parameters. See the UHF Parameters section for parameter definitions and value types.

## 5.2.16. Set Application Parameters

|                                                                                                                                                                                                                                                                        | Type                  | Description                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------|
| <code>device_status_t device_set_app_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value, bool save)</code><br><code>device_status_t device_save_app_param(device_handle_t dev, uint8_t id, const device_param_value_t* param_value)</code> |                       |                                                               |
| Parameter                                                                                                                                                                                                                                                              |                       |                                                               |
| dev                                                                                                                                                                                                                                                                    | device_handle_t       | Device handle                                                 |
| id                                                                                                                                                                                                                                                                     | uint8_t               | Application parameter type                                    |
| param_val                                                                                                                                                                                                                                                              | device_param_value_t* | Parameter object used to store the parameter value to be set. |
| saved                                                                                                                                                                                                                                                                  | bool                  | Whether to persist (save to device Flash)                     |
| Return Value                                                                                                                                                                                                                                                           |                       |                                                               |
|                                                                                                                                                                                                                                                                        | device_status_t       | Returns the set result                                        |
| Description                                                                                                                                                                                                                                                            |                       |                                                               |
| Set application parameters. See the Application Parameters section for parameter definitions and value types.                                                                                                                                                          |                       |                                                               |

## 5.2.17. Query GPIO Mode

|                                                                                                                       | Type                | Description                                                  |
|-----------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------|
| <code>device_status_t device_get_gpio_mode(device_handle_t dev, device_gpio_mode_t* modes, uint32_t* out_mask)</code> |                     |                                                              |
| Parameter                                                                                                             |                     |                                                              |
| dev                                                                                                                   | device_handle_t     | Device handle                                                |
| modes                                                                                                                 | device_gpio_mode_t* | Used to store queried GPIO mode results, stored as an array. |
| out_mask                                                                                                              | uint32_t*           | Used to store the queried GPIO mask, 1 indicates the         |

|                              |                 |                          |
|------------------------------|-----------------|--------------------------|
|                              |                 | GPIO exists.             |
| Return Value                 |                 |                          |
|                              | device_status_t | Returns the query result |
| Description                  |                 |                          |
| Query the reader GPIO modes. |                 |                          |

## 5.2.18. Set GPIO Mode

|                                                                                                                 | Type                | Description                                                |
|-----------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------|
| device_status_t device_set_gpio_mode(device_handle_t dev,<br>uint32_t mask,<br>const device_gpio_mode_t* modes) |                     |                                                            |
| Parameter                                                                                                       |                     |                                                            |
| dev                                                                                                             | device_handle_t     | Device handle                                              |
| mask                                                                                                            | uint32_t*           | Mask of GPIOs to configure.                                |
| modes                                                                                                           | device_gpio_mode_t* | Used to store GPIO modes to configure, stored as an array. |
| Return Value                                                                                                    |                     |                                                            |
|                                                                                                                 | device_status_t     | Returns the query result                                   |
| Description                                                                                                     |                     |                                                            |
| Set the device GPIO modes.                                                                                      |                     |                                                            |

## 5.2.19. Query GPIO Logic Level State

|                                                                                                        | Type            | Description                                                       |
|--------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------|
| device_status_t device_get_gpio_state(device_handle_t dev,<br>uint32_t* out_state, uint32_t* out_mask) |                 |                                                                   |
| Parameter                                                                                              |                 |                                                                   |
| dev                                                                                                    | device_handle_t | Device handle                                                     |
| out_state                                                                                              | uint32_t*       | Used to store the queried GPIO state mask.                        |
| out_mask                                                                                               | uint32_t*       | Used to store the queried GPIO mask, 1 indicates the GPIO exists. |
| Return Value                                                                                           |                 |                                                                   |
|                                                                                                        | device_status_t | Returns the query result                                          |



|                                                                               |
|-------------------------------------------------------------------------------|
| Description                                                                   |
| Query device GPIO states. Output as a bitmask: 1 = high level, 0 = low level. |

### 5.2.20. Set GPIO Logic Level State

|                                                                                           | Type            | Description                               |
|-------------------------------------------------------------------------------------------|-----------------|-------------------------------------------|
| device_status_t device_set_gpio_state(device_handle_t dev, uint32_t mask, uint32_t state) |                 |                                           |
| Parameter                                                                                 |                 |                                           |
| dev                                                                                       | device_handle_t | Device handle                             |
| mask                                                                                      | uint32_t*       | Mask of GPIOs to configure.               |
| state                                                                                     | uint32_t*       | Used to store the GPIO level mask to set. |
| Return Value                                                                              |                 |                                           |
|                                                                                           | device_status_t | Returns the query result                  |
| Description                                                                               |                 |                                           |
| Set device GPIO states. Input as a bitmask: 1 = high level, 0 = low level.                |                 |                                           |

### 5.2.21. Buzzer Beep

|                                                                        | Type            | Description                            |
|------------------------------------------------------------------------|-----------------|----------------------------------------|
| device_status_t device_beep(device_handle_t dev, uint32_t duration_ms) |                 |                                        |
| Parameter                                                              |                 |                                        |
| dev                                                                    | device_handle_t | Device handle                          |
| duration_ms                                                            | uint32_t        | Buzzer beep duration, in milliseconds. |
| Return Value                                                           |                 |                                        |
|                                                                        | device_status_t | Returns the set result                 |
| Description                                                            |                 |                                        |
| Trigger a single buzzer beep.                                          |                 |                                        |

### 5.2.22. Start Asynchronous RFID Inventory

|                                                                                    | Type | Description |
|------------------------------------------------------------------------------------|------|-------------|
| device_status_t device_rfid_inventory(device_handle_t dev, const rfid_inventory_t* |      |             |

|                                                                                                                                                         |                   |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------|
| inventory)                                                                                                                                              |                   |                                           |
| Parameter                                                                                                                                               |                   |                                           |
| dev                                                                                                                                                     | device_handle_t   | Device handle                             |
| inventory                                                                                                                                               | rfid_inventory_t* | Inventory parameters                      |
| Return Value                                                                                                                                            |                   |                                           |
|                                                                                                                                                         | device_status_t   | Returns the async inventory start result. |
| Description                                                                                                                                             |                   |                                           |
| The device performs asynchronous RFID inventory. After starting successfully, inventoried tags are passed to the application via the RFID tag callback. |                   |                                           |

### 5.2.23. Stop RFID

|                                                                                        | Type            | Description             |
|----------------------------------------------------------------------------------------|-----------------|-------------------------|
| device_status_t device_rfid_stop(device_handle_t dev)                                  |                 |                         |
| Parameter                                                                              |                 |                         |
| dev                                                                                    | device_handle_t | Device handle           |
| Return Value                                                                           |                 |                         |
|                                                                                        | device_status_t | Returns the stop result |
| Description                                                                            |                 |                         |
| Used to stop RFID operations. Commonly used for manual stop of asynchronous inventory. |                 |                         |

### 5.2.24. Read RFID Tag Data

|                                                                              | Type            | Description                                                         |
|------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------|
| device_status_t device_rfid_read_tag(device_handle_t dev, rfid_read_t* read) |                 |                                                                     |
| Parameter                                                                    |                 |                                                                     |
| dev                                                                          | device_handle_t | Device handle                                                       |
| read                                                                         | rfid_read_t     | Read RFID tag parameters, passes read parameters and returned data. |
| Return Value                                                                 |                 |                                                                     |
|                                                                              | device_status_t | Returns the RFID tag read result.                                   |

|                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                                                                                                                                                                                                                                                       |
| Read RFID tag with specified parameters. After reading, the data is returned through the 'read' parameter.<br>When operating on a specific tag via tag operation API, first call the Set UHF Parameter API to configure TAG_SELECTn for tag matching, then invoke the tag operation API. |

### 5.2.25. Write RFID Tag Data

|                                                                                                                                                                                                                         | Type            | Description                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------|
| device_status_t device_rfid_write_tag(device_handle_t dev,<br>const rfid_write_t* write)                                                                                                                                |                 |                                                                       |
| <b>Parameter</b>                                                                                                                                                                                                        |                 |                                                                       |
| dev                                                                                                                                                                                                                     | device_handle_t | Device handle                                                         |
| write                                                                                                                                                                                                                   | rfid_write_t*   | Write RFID tag parameters, passes write parameters and data to write. |
| <b>Return Value</b>                                                                                                                                                                                                     |                 |                                                                       |
|                                                                                                                                                                                                                         | device_status_t | Returns the RFID tag write result.                                    |
| <b>Description</b>                                                                                                                                                                                                      |                 |                                                                       |
| Write RFID tag with specified parameters.<br>When operating on a specific tag via tag operation API, first call the Set UHF Parameter API to configure TAG_SELECTn for tag matching, then invoke the tag operation API. |                 |                                                                       |

### 5.2.26. Lock RFID Tag

|                                                                                       | Type            | Description              |
|---------------------------------------------------------------------------------------|-----------------|--------------------------|
| device_status_t device_rfid_lock_tag(device_handle_t dev,<br>const rfid_lock_t* lock) |                 |                          |
| <b>Parameter</b>                                                                      |                 |                          |
| dev                                                                                   | device_handle_t | Device handle            |
| lock                                                                                  | rfid_lock_t     | RFID tag lock parameters |
| <b>Return Value</b>                                                                   |                 |                          |

|                                                                                                                                                                                                                                   |                 |                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|
|                                                                                                                                                                                                                                   | device_status_t | Returns the lock tag result. |
| <b>Description</b>                                                                                                                                                                                                                |                 |                              |
| <p>Lock RFID tag with specified parameters.</p> <p>When operating on a specific tag via tag operation API, first call the Set UHF Parameter API to configure TAG_SELECTn for tag matching, then invoke the tag operation API.</p> |                 |                              |

## 5.2.27. Kill RFID Tag

|                                                                                                                                                                                                                            | Type            | Description                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|
| <code>device_status_t device_rfid_kill_tag(device_handle_t dev, const rfid_kill_t* kill)</code>                                                                                                                            |                 |                              |
| <b>Parameter</b>                                                                                                                                                                                                           |                 |                              |
| dev                                                                                                                                                                                                                        | device_handle_t | Device handle                |
| kill                                                                                                                                                                                                                       | rfid_kill_t     | RFID tag kill parameters     |
| <b>Return Value</b>                                                                                                                                                                                                        |                 |                              |
|                                                                                                                                                                                                                            | device_status_t | Returns the kill tag result. |
| <b>Description</b>                                                                                                                                                                                                         |                 |                              |
| <p>Kill RFID tag with specified parameters. When operating on a specific tag via tag operation API, first call the Set UHF Parameter API to configure TAG_SELECTn for tag matching, then invoke the tag operation API.</p> |                 |                              |

## 6. Type Definitions

### 6.1. Enumerations

#### 6.1.1. device\_gpio\_index\_t

GPIO index, used for GPIO operations, GPIO triggers, etc.

| Definition | Description |
|------------|-------------|
|------------|-------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| DEVICE_GPIO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO1 index |
| DEVICE_GPIO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO2 index |
| DEVICE_GPIO3<br><i>[<a href="#">↗</a> ISSUE:<br/>           DEVICE_GPIO3<br/>           appears twice in this<br/>           enum. The second<br/>           occurrence should<br/>           likely be<br/>           DEVICE_GPIO4. This<br/>           creates a gap in GPIO<br/>           numbering (GPIO1,<br/>           GPIO2, GPIO3, GPIO3,<br/>           GPIO4, GPIO5, GPIO6,<br/>           GPIO7 — missing<br/>           unique<br/>           DEVICE_GPIO4).]</i> | GPIO3 index |
| DEVICE_GPIO3<br><i>[<a href="#">↗</a> ISSUE:<br/>           DEVICE_GPIO3<br/>           appears twice in this<br/>           enum. The second<br/>           occurrence should<br/>           likely be<br/>           DEVICE_GPIO4. This<br/>           creates a gap in GPIO<br/>           numbering (GPIO1,<br/>           GPIO2, GPIO3, GPIO3,<br/>           GPIO4, GPIO5, GPIO6,<br/>           GPIO7 — missing<br/>           unique<br/>           DEVICE_GPIO4).]</i> | GPIO3 index |
| DEVICE_GPIO4                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO4 index |
| DEVICE_GPIO5                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO5 index |
| DEVICE_GPIO6                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO6 index |
| DEVICE_GPIO7                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO7 index |

### 6.1.2. device\_gpio\_mode\_t

GPIO mode

| Definition         | Description      |
|--------------------|------------------|
| DEVICE_GPIO_OUTPUT | GPIO output mode |
| DEVICE_GPIO_INPUT  | GPIO input mode  |

### 6.1.3. device\_status\_t

Device status code

| Definition                               | Description                            |
|------------------------------------------|----------------------------------------|
| DEVICE_OK                                | Success                                |
| DEVICE_ERR_CMD_INVALID_TYPE              | Invalid command type                   |
| DEVICE_ERR_CMD_INVALID_LEN               | Invalid command length                 |
| DEVICE_ERR_CMD_INVALID_PARAM             | Invalid command parameter              |
| DEVICE_ERR_CMD_EXEC                      | Command execution error                |
| DEVICE_ERR_NO_RESPONSE                   | Command no response                    |
| DEVICE_ERR_GPIO_OVER_RANGE               | GPIO index out of range                |
| DEVICE_ERR_GPIO_INVALID_MODE             | Invalid GPIO mode                      |
| DEVICE_ERR_GPIO_OTHER_ERROR              | Other GPIO error                       |
| DEVICE_ERR_PARAM_INVALID_TYPE            | Invalid parameter type                 |
| DEVICE_ERR_PARAM_INVALID_VAL             | Invalid parameter value                |
| DEVICE_ERR_PARAM_INVALID_LENGTH          | Invalid parameter length               |
| DEVICE_ERR_PARAM_GET_FAILED              | Failed to get parameter                |
| DEVICE_ERR_PARAM_SET_FAILED              | Failed to set parameter                |
| DEVICE_ERR_PARAM_SAVE_FAILED             | Failed to save parameter               |
| DEVICE_ERR_PARAM_OTHER_ERROR             | Other parameter error                  |
| DEVICE_ERR_TAG_OTHER_ERROR               | Tag returned other error               |
| DEVICE_ERR_TAG_NOT_SUPPORT               | Tag returned not supported             |
| DEVICE_ERR_TAG_INSUFFICIENT_PRIVILEGES   | Tag returned insufficient privileges   |
| DEVICE_ERR_TAG_MEMORY_OVERRUN            | Tag returned memory overrun            |
| DEVICE_ERR_TAG_MEMORY_LOCKED             | Tag returned memory locked             |
| DEVICE_ERR_TAG_CRYPTOGRAPHIC_SUITE_ERROR | Tag returned cryptographic suite error |
| DEVICE_ERR_TAG_COMMAND_NOT_ENCAPSULATED  | Tag returned command not encapsulated  |
| DEVICE_ERR_TAG_RESPONSEBUFFER_OVERFLOW   | Tag returned response buffer overflow  |
| DEVICE_ERR_TAG_SECURITY_TIMEOUT          | Tag returned security timeout          |
| DEVICE_ERR_TAG_INSUFFICIENT_POWER        | Tag returned insufficient power        |

| Definition                                                                                | Description            |
|-------------------------------------------------------------------------------------------|------------------------|
| DEVICE_ERR_TAG_LOST                                                                       | Tag lost               |
| DEVICE_ERR_TAG_NON_SPECIFIC_ERROR                                                         | Non-specific error     |
| DEVICE_ERR_RFID_INVALID_REGION                                                            | Invalid RFID region    |
| DEVICE_ERR_RFID_INVALID_VALUE                                                             | Invalid RFID value     |
| DEVICE_ERR_RFID_EXEC                                                                      | RFID execution error   |
| DEVICE_ERR_RFID_NO_ACK                                                                    | RFID no acknowledgment |
| DEVICE_ERR_RFID_INTERNAL_ERROR<br>[ ⚠ TYPO: 'INTERNEL_ERROR' should be 'INTERNAL_ERROR'.] | RFID internal error    |
| DEVICE_ERR_RFID_OTHER_ERROR                                                               | Other RFID error       |
| DEVICE_ERR_BUSY                                                                           | Device busy            |
| DEVICE_ERR_INTERNAL                                                                       | Internal error         |
| DEVICE_ERR_INVALID_LINK                                                                   | Invalid connection     |

#### 6.1.4. antenna\_index\_t

Antenna index, used for carrier test, tag data, etc.

| Definition | Description |
|------------|-------------|
| ANTENNA_1  | Antenna 1   |
| ANTENNA_2  | Antenna 2   |
| ...        | ...         |
| ANTENNA_16 | Antenna 16  |

#### 6.1.5. antenna\_mask\_t

Antenna mask, used for antenna configuration

| Definition      | Description |
|-----------------|-------------|
| ANTENNA_MASK_1  | Antenna 1   |
| ANTENNA_MASK_2  | Antenna 2   |
| ...             | ...         |
| ANTENNA_MASK_16 | Antenna 16  |



## 6.1.6. uhf\_bank\_type\_t

EPC tag memory banks, used for tag read/write operations

| Definition        | Description                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UHF_BANK_RESERVED | Gen2 Reserved Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i> |
| UHF_BANK_EPC      | Gen2 EPC Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i>      |
| UHF_BANK_TID      | Gen2 TID Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i>      |
| UHF_BANK_USER     | Gen2 User Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i>     |

## 6.1.7. uhf\_lock\_bank\_type\_t

UHF tag lock banks, used for tag lock operations

| Definition         | Description                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UHF_LOCK_BANK_USER | Gen2 User Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i> |
| UHF_LOCK_BANK_TID  | Gen2 TID Bank<br><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2'</i>                                                           |

|                         |                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | or 'ISO 18000-6C' for clarity in the SDK documentation.]                                                                                                                                     |
| UHF_LOCK_BANK_EPC       | <b>Gen2 EPC Bank</b><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i>        |
| UHF_LOCK_BANK_ACCESSPWD | <b>Gen2 Access Password</b><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i> |
| UHF_LOCK_BANK_KILLPWD   | <b>Gen2 Kill Password</b><br><i>[⚠ CLARITY: '6C' is shorthand for ISO/IEC 18000-6C (EPC Class 1 Gen 2). Consider using 'Gen2' or 'ISO 18000-6C' for clarity in the SDK documentation.]</i>   |

### 6.1.8. rfid\_lock\_type\_t

UHF tag lock type, used for tag lock operations

| Definition                       | Description              |
|----------------------------------|--------------------------|
| RFID_LOCK_TYPE_WRITEABLE         | Read and write           |
| RFID_LOCK_TYPE_WRITE_ALWAYS      | Permanently writable     |
| RFID_LOCK_TYPE_WRITE_WITH_PASSWD | Password-protected write |
| RFID_LOCK_TYPE_NOT_WRITEABLE     | Permanently non-writable |

## 6.2. Structures

## 6.2.1. rfid\_inventory\_t

RFID inventory parameters

| Properties       | Type     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask             | uint32_t | Inventory options, each bit corresponds to a function:<br><br>bit0: cache tags (do not transmit tag data)<br>bit1: request TID transmission<br>bit2: request frequency transmission<br>bit3: request phase transmission<br>bit4: request read count transmission<br>bit5: request timestamp transmission<br>bit6: request RSSI transmission<br><br>bit8: enable inventory timeout option<br>bit9: enable tag select option<br>bit10: enable password option<br>bit11: enable preset password option<br><br>bit16: enable temperature tag option<br>bit17: enable humidity tag option<br>bit18: enable LED tag option<br>bit19: enable Impinj tag option<br>bit20: enable NXP tag option.<br><br>bit24: enable tag read<br><br>bit29: discard tags with RSSI below specified value<br>bit30: merge and report |
| timeout          | uint32_t | Inventory duration, in ms. 0 means continuous inventory until StopRFID command. Default is 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| select_en        | uint32_t | Tag select enable mask: bit0 = select 1, bit1 = select 2, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| passwd           | uint32_t | Access password                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| preset_passwd    | uint32_t | Preset password index                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| temperature.type | uint32_t | Temperature tag type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                                                                        |          |                                                                                                             |
|--------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------|
| temperature.args                                                                                       | uint32_t | Temperature tag parameters                                                                                  |
| humidity.type<br><br><i>[⚠️ TYPO: 'humility' should be 'humidity' in the rfid_inventory_t struct.]</i> | uint32_t | Humidity tag type                                                                                           |
| humidity.args<br><br><i>[⚠️ TYPO: 'humility' should be 'humidity' in the rfid_inventory_t struct.]</i> | uint32_t | Humidity tag parameters                                                                                     |
| led.type                                                                                               | uint32_t | Temperature LED type                                                                                        |
| led.args                                                                                               | uint32_t | Temperature LED parameters                                                                                  |
| impinj<br><br><i>[⚠️ TYPO: 'ImpinJ' should be 'Impinj' (the company name).]</i>                        | uint32_t | Impinj Tag Options<br>bit0: enable TagFocus<br>bit1: enable FastID<br>bit8: enable Gen2s (extended Q range) |
| nxp                                                                                                    | uint32_t | NXP tag options                                                                                             |
| rss_i_threshold                                                                                        | int16_t  | RSSI filter threshold, in 0.01 dBm units.                                                                   |
| merge_time                                                                                             | uint32_t | Tag merge/deduplication time, in ms.                                                                        |
| read.bank                                                                                              | uint16_t | Tag read memory bank                                                                                        |
| read.wpoint                                                                                            | uint16_t | Tag read word address                                                                                       |
| read.wlength                                                                                           | uint16_t | Tag read word length                                                                                        |

## 6.2.2. rfid\_kill\_t

RFID tag kill parameters

| Properties | Type     | Description                                                                                                                                                                                      |
|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask       | uint32_t | Kill options, each bit corresponds to a function:<br>bit8: enable kill timeout option<br>bit9: enable tag select option<br>bit10: enable password option<br>bit11: enable preset password option |
| timeout    | uint32_t | Kill timeout, in ms.                                                                                                                                                                             |
| select_en  | uint32_t | Tag select enable mask: bit0 = select 1, bit1 = select 2, etc.                                                                                                                                   |

|               |          |                       |
|---------------|----------|-----------------------|
| passwd        | uint32_t | Access password       |
| preset_passwd | uint32_t | Preset password index |

### 6.2.3. rfid\_lock\_t

RFID tag lock parameters

| Properties    | Type     | Description                                                                                                                                                                                      |
|---------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask          | uint32_t | Lock options, each bit corresponds to a function:<br>bit8: enable lock timeout option<br>bit9: enable tag select option<br>bit10: enable password option<br>bit11: enable preset password option |
| timeout       | uint32_t | Lock timeout, in ms.                                                                                                                                                                             |
| select_en     | uint32_t | Tag select enable mask: bit0 = select 1, bit1 = select 2, etc.                                                                                                                                   |
| passwd        | uint32_t | Access password                                                                                                                                                                                  |
| preset_passwd | uint32_t | Preset password index                                                                                                                                                                            |
| bank          | uint16_t | Lock memory bank                                                                                                                                                                                 |
| type          | uint16_t | Lock action type                                                                                                                                                                                 |

### 6.2.4. rfid\_read\_t

Tag read parameters

| Properties | Type     | Description                                                                                                                                                                                      |
|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask       | uint32_t | Read options, each bit corresponds to a function:<br>bit8: enable read timeout option<br>bit9: enable tag select option<br>bit10: enable password option<br>bit11: enable preset password option |
| timeout    | uint32_t | Read timeout, in ms.                                                                                                                                                                             |
| select_en  | uint32_t | Tag select enable mask: bit0 = select 1, bit1 = select 2, etc.                                                                                                                                   |

|               |           |                       |
|---------------|-----------|-----------------------|
| passwd        | uint32_t  | Access password       |
| preset_passwd | uint32_t  | Preset password index |
| bank          | uint16_t  | Read memory bank      |
| wpoint        | uint16_t  | Read word address     |
| wlength       | uint16_t  | Read word length      |
| data          | uint8_t[] | Read data             |

### 6.2.5. rfid\_write\_t

RFID tag write parameters

| Properties    | Type      | Description                                                                                                                                                                                        |
|---------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mask          | uint32_t  | Write options, each bit corresponds to a function:<br>bit8: enable write timeout option<br>bit9: enable tag select option<br>bit10: enable password option<br>bit11: enable preset password option |
| timeout       | uint32_t  | Write timeout, in ms.                                                                                                                                                                              |
| select_en     | uint32_t  | Tag select enable mask: bit0 = select 1, bit1 = select 2, etc.                                                                                                                                     |
| passwd        | uint32_t  | Access password                                                                                                                                                                                    |
| preset_passwd | uint32_t  | Preset password index                                                                                                                                                                              |
| wblock        | uint16_t  | Block size                                                                                                                                                                                         |
| bank          | uint16_t  | Write memory bank                                                                                                                                                                                  |
| wpoint        | uint16_t  | Write word address                                                                                                                                                                                 |
| wlength       | uint16_t  | Write word length                                                                                                                                                                                  |
| data          | uint8_t[] | Data to write                                                                                                                                                                                      |

### 6.2.6. device\_rfid\_tag\_t

RFID tag data

| Properties | Type     | Description                 |
|------------|----------|-----------------------------|
| type       | uint8_t  | Tag type                    |
| pc         | uint16_t | PC (Protocol Control) field |

|                                                                                                                                                                                                                                                                                                                   |           |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------|
| epc                                                                                                                                                                                                                                                                                                               | uint8_t[] | EPC data                                |
| epc_len                                                                                                                                                                                                                                                                                                           | uint8_t   | EPC data length                         |
| tid                                                                                                                                                                                                                                                                                                               | uint8_t[] | TID data                                |
| tid_len                                                                                                                                                                                                                                                                                                           | uint8_t   | TID data length                         |
| data                                                                                                                                                                                                                                                                                                              | uint8_t[] | Additional data read during inventory   |
| data_len                                                                                                                                                                                                                                                                                                          | uint8_t   | Additional data length                  |
| antenna                                                                                                                                                                                                                                                                                                           | uint8_t   | Antenna index that detected the tag     |
| count                                                                                                                                                                                                                                                                                                             | uint32_t  | Read count                              |
| rss                                                                                                                                                                                                                                                                                                               | int16_t   | Tag RSSI value, in 0.01 dBm units.      |
| frequency_khz                                                                                                                                                                                                                                                                                                     | uint32_t  | Tag read frequency, in kHz.             |
| phase                                                                                                                                                                                                                                                                                                             | int16_t   | Tag read phase, in 0.1° units.          |
| timestamp_sec                                                                                                                                                                                                                                                                                                     | uint32    | UTC timestamp, seconds part.            |
| timestamp_ms                                                                                                                                                                                                                                                                                                      | uint32    | UTC timestamp, milliseconds part.       |
| <p></p> <p><i>INCONSISTENCY: Field 'timestamp_ms' is described as 'UTC timestamp, nanoseconds part' (nanoseconds), but the '_ms' suffix suggests milliseconds. Verify the actual unit — this mismatch could cause bugs.]</i></p> |           |                                         |
| temperature                                                                                                                                                                                                                                                                                                       | int16_t   | Tag temperature value, in 0.01°C units. |
| humidity                                                                                                                                                                                                                                                                                                          | int16_t   | Tag humidity value, in 0.01% units.     |

## Appendix 1: Parameter Table

### 1.1 Device Parameters

| Parameter ID | Parameter Name | R/W | Value Description       |
|--------------|----------------|-----|-------------------------|
| 0x00         | System Version | RO  | byte[4]: version number |

|      |                             |    |                                                     |
|------|-----------------------------|----|-----------------------------------------------------|
|      |                             |    | string: version description                         |
| 0x01 | Motherboard Version         | RO | byte[4]: version number                             |
|      |                             |    | string: version description                         |
| 0x02 | Motherboard Serial Number   | RO | string: serial number                               |
| 0x03 | Ethernet MAC                | RO | byte[6]: MAC address                                |
| 0x04 | WIFI MAC                    | RO | byte[6]: MAC address                                |
| 0x05 | BT_MAC                      | RO | byte[6]: MAC address                                |
| 0x0B | Wi-Fi Firmware Version      | RO | string: version description                         |
| 0x10 | Device Name                 | RW | string: device description                          |
| 0x11 | Asset Number                | RW | string: asset number                                |
| 0x12 | Device Date/Time            | RW | int: UTC seconds                                    |
|      |                             |    | int: UTC nanoseconds                                |
| 0x13 | Device Boot Info            | RO | int: device uptime (seconds)                        |
|      |                             |    | int: boot reason code                               |
| 0x14 | Power-on Event              | RW | bool: power-on event enable                         |
|      |                             |    | array: commands (full commands, multiple supported) |
| 0x1E | Device Temperature          | RO | short: RFID module temperature (0.01°C)             |
| 0x1F | Device Battery SoC          | RO | short: battery SoC (0.01%)                          |
| 0x20 | Device Address              | RW | byte: address, 1~250                                |
| 0x21 | RS232 Properties            | RW | int: baud rate                                      |
|      |                             |    | byte: data bits, must be 8                          |
|      |                             |    | byte: stop bits, can be 0, 1, 2                     |
|      |                             |    | byte: parity, 0=None, 1=Odd, 2=Even                 |
|      |                             |    | byte: reserved, fill with 0                         |
| 0x22 | RS485 Properties            | RW | int: baud rate                                      |
|      |                             |    | byte: data bits, must be 8                          |
|      |                             |    | byte: stop bits, can be 0, 1, 2                     |
|      |                             |    | byte: parity, 0=None, 1=Odd, 2=Even                 |
|      |                             |    | byte: reserved, fill with 0                         |
| 0x24 | Ethernet Network Properties | RW | bool: DHCP enable                                   |
|      |                             |    | byte[4]: IP address                                 |
|      |                             |    | byte[4]: subnet mask                                |
|      |                             |    | byte[4]: gateway                                    |
|      |                             |    | byte[4]: DNS                                        |



|                                                                                                                                                                                                |                          |    |                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----|--------------------------------------------------------------------------------|
|                                                                                                                                                                                                |                          |    | byte: mode, 0=TCP Server, 1=TCP Client, 2=UDP Server, 3=UDP Client             |
|                                                                                                                                                                                                |                          |    | short: local port                                                              |
|                                                                                                                                                                                                |                          |    | string: remote address                                                         |
|                                                                                                                                                                                                |                          |    | short: remote port                                                             |
| 0x25                                                                                                                                                                                           | Wi-Fi Network Properties | RW | bool: DHCP enable                                                              |
|                                                                                                                                                                                                |                          |    | byte[4]: IP address                                                            |
|                                                                                                                                                                                                |                          |    | byte[4]: subnet mask                                                           |
|                                                                                                                                                                                                |                          |    | byte[4]: gateway                                                               |
|                                                                                                                                                                                                |                          |    | byte[4]: DNS                                                                   |
|                                                                                                                                                                                                |                          |    | byte: mode, 0=TCP Server, 1=TCP Client, 2=UDP Server, 3=UDP Client             |
|                                                                                                                                                                                                |                          |    | short: local port                                                              |
| 0x26                                                                                                                                                                                           | Wi-Fi AP Connection      | RW | string: SSID                                                                   |
|                                                                                                                                                                                                |                          |    | string: password                                                               |
| 0x27                                                                                                                                                                                           | Wi-Fi Connection Info    | RO | string: SSID                                                                   |
|                                                                                                                                                                                                |                          |    | bool: connection status                                                        |
| 0x2F<br><br><i>ISSUE: Default connection parameter 0x2F lists '0:RS232 1:RS485 2:RS232' — index 0 and 2</i> | Default Connection       | RW | byte: default connection, 0=RS232, 1=RS485, 2=RS232, 3=Ethernet, 4=Wi-Fi, 5=BT |

|                                                                                     |                               |    |                                                                    |
|-------------------------------------------------------------------------------------|-------------------------------|----|--------------------------------------------------------------------|
| <i>both map to RS232. Index 2 should likely be Bluetooth or another interface.]</i> |                               |    |                                                                    |
| 0x51                                                                                | Buzzer Parameters             | RW | bool: buzzer enable                                                |
|                                                                                     |                               |    | short: buzzer on duration                                          |
|                                                                                     |                               |    | short: buzzer off duration                                         |
| 0x52                                                                                | Comm Indicator LED Parameters | RW | bool: comm LED enable                                              |
|                                                                                     |                               |    | short: comm LED on duration                                        |
|                                                                                     |                               |    | short: comm LED off duration                                       |
| 0x53                                                                                | Tag Indicator LED Parameters  | RW | bool: tag LED enable                                               |
|                                                                                     |                               |    | short: tag LED on duration                                         |
|                                                                                     |                               |    | short: tag LED off duration                                        |
| 0x60                                                                                | GPIO1 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|                                                                                     |                               |    | array: commands (full commands, multiple supported)                |
| 0x61                                                                                | GPIO2 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|                                                                                     |                               |    | array: commands (full commands, multiple supported)                |
| 0x62                                                                                | GPIO3 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |

|      |                               |    |                                                                    |
|------|-------------------------------|----|--------------------------------------------------------------------|
|      |                               |    | array: commands (full commands, multiple supported)                |
| 0x63 | GPIO4 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|      |                               |    | array: commands (full commands, multiple supported)                |
| 0x64 | GPIO5 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|      |                               |    | array: commands (full commands, multiple supported)                |
| 0x65 | GPIO6 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|      |                               |    | array: commands (full commands, multiple supported)                |
| 0x66 | GPIO7 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|      |                               |    | array: commands (full commands, multiple supported)                |
| 0x67 | GPIO8 Interrupt Trigger Event | RW | byte: edge type, 0=none, 1=rising edge, 2=falling edge, 3=any edge |
|      |                               |    | array: commands (full commands, multiple supported)                |

## 1.2 RFID Parameters

| Parameter ID | Parameter Name        | R/W | Value Description                                                                                                                                                                                                                                                     |
|--------------|-----------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x01         | RFID Version          | RO  | byte[4]: version number                                                                                                                                                                                                                                               |
|              |                       |     | string: version description                                                                                                                                                                                                                                           |
| 0x02         | RFID SN               | RO  | string: serial number<br><i>[⚠️ TYPO: 'stirng' should be 'string' in the value description.]</i>                                                                                                                                                                      |
| 0x0F         | RFID Properties       | RO  | int: supported protocol bitmask, bit0: ISO/IEC 18000-6B, bit1: ISO/IEC 18000-6C/EPC C1G2, bit2: ISO/IEC 156938000-6D bit4: GB29768 bit5: GJB7377.1 bit16: ISO14443A bit17: ISO14443B bit18: ISO15693                                                                  |
|              |                       |     | byte: number of antennas                                                                                                                                                                                                                                              |
|              |                       |     | short: max power (0.01 dBm)                                                                                                                                                                                                                                           |
|              |                       |     | short: min power (0.01 dBm)                                                                                                                                                                                                                                           |
|              |                       |     | short: default power (0.01 dBm)                                                                                                                                                                                                                                       |
|              |                       |     | short: power step (0.01 dBm)                                                                                                                                                                                                                                          |
| 0x10         | RFID Current Protocol | RW  | int: working protocol bitmask. Device may support single or multiple simultaneous protocols for inventory. bit0: ISO/IEC 18000-6B, bit1: ISO/IEC 18000-6C./EPC C1G2 bit2: ISO18000-6D bit4: GB29768 bit5: GJB7377.1 bit16: ISO14443A bit17: ISO14443B bit18: ISO15693 |
| 0x1E         | RFID Temperature      | RO  | short: RFID module temperature (0.01°C)                                                                                                                                                                                                                               |
| 0x20         | RFID Idle Mode        | RW  | bool: enable                                                                                                                                                                                                                                                          |
|              |                       |     | short: cycle (ms)                                                                                                                                                                                                                                                     |
|              |                       |     | byte: idle ratio, 0~100%                                                                                                                                                                                                                                              |

|      |                 |    |                                                                                     |
|------|-----------------|----|-------------------------------------------------------------------------------------|
| 0x50 | RFID EAS Event  | RW | byte: mode, 0=disable EAS, 1=enable EAS, 2=enable EAS with auto-read on power-on    |
|      |                 |    | short: data source, ISO/IEC 18000-6C: 0=EPC, 1=TID.                                 |
|      |                 |    | short: start bit address, -1 = match from end                                       |
|      |                 |    | array: mask                                                                         |
|      |                 |    | array: expected value                                                               |
|      |                 |    | array: commands on match (full commands, multiple supported)                        |
|      |                 |    | array: commands on no match (full commands, multiple supported)                     |
| 0x5F | Whitelist Event | RW | bool: enable                                                                        |
|      |                 |    | short: data source, ISO/IEC 18000-6C: 0=EPC, 1=TID; other protocols: 0=tag UID.     |
|      |                 |    | short: start bit address                                                            |
|      |                 |    | short: data bit length, -1 = to end                                                 |
|      |                 |    | byte: format, 0=raw, 1=hex string                                                   |
|      |                 |    | array: commands on match (full commands, multiple supported)                        |
| 0x60 | Wiegand Output  | RW | byte: mode, 0=disable Wiegand output, 1=enable, 2=enable with auto-read on power-on |
|      |                 |    | short: data source, ISO/IEC 18000-6C: 0=EPC, 1=TID; other protocols: 0=tag UID.     |
|      |                 |    | short: start bit address, -1 = send from end                                        |

|      |                                 |    |                                                                                                               |
|------|---------------------------------|----|---------------------------------------------------------------------------------------------------------------|
|      |                                 |    | byte: data bit length                                                                                         |
| 0x61 | Custom<br>Format<br>Output      | RW | byte: mode, 0=disable custom format output, 1=enable, 2=enable with auto-read on power-on                     |
|      |                                 |    | short: data source, ISO/IEC 18000-6C: 0=EPC, 1=TID; other protocols: 0=tag UID.                               |
|      |                                 |    | short: start bit address, -1 = send end data                                                                  |
|      |                                 |    | short: data bit length                                                                                        |
|      |                                 |    | byte: output format, 0=raw, 1=hex string                                                                      |
|      |                                 |    | byte: checksum method                                                                                         |
|      |                                 |    | int: additional data mask, bit0: antenna, bit5: RSSI, bit6: frequency, bit7: phase, bit29: SN, bit30: ...etID |
|      |                                 |    | array: frame header                                                                                           |
|      |                                 |    | array: frame footer                                                                                           |
|      |                                 |    | array: additional data delimiter                                                                              |
| 0x62 | Keyboard<br>Emulation<br>Output | RW | byte: mode, 0=disable keyboard emulation, 1=enable, 2=enable with auto-read on power-on                       |
|      |                                 |    | short: data source, ISO/IEC 18000-6C: 0=EPC, 1=TID; other protocols: 0=tag UID.                               |

|      |                           |    |                                                                                 |
|------|---------------------------|----|---------------------------------------------------------------------------------|
|      |                           |    | byte: output format, 0=raw, 1=hex string                                        |
|      |                           |    | array: frame footer                                                             |
| 0x80 | RFID Antenna Status       | RO | int: antenna connection mask, bit0=antenna 1 status, bit1=antenna 2 status, ... |
| 0x81 | Antenna Configuration     | RW | int: working antenna mask, bit0=antenna 1, bit1=antenna 2, ...                  |
|      |                           |    | short: max antenna dwell time (ms)                                              |
| 0x82 | All Antennas Power Config | WO | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x90 | Antenna 1 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x91 | Antenna 2 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x92 | Antenna 3 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x93 | Antenna 4 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x94 | Antenna 5 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x95 | Antenna 6 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x96 | Antenna 7 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x97 | Antenna 8 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x98 | Antenna 9 Power           | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x99 | Antenna 10 Power          | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x9A | Antenna 11 Power          | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x9B | Antenna 12 Power          | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |
| 0x9C | Antenna 13 Power          | RW | short: read power (0.01 dBm)                                                    |
|      |                           |    | short: write power (0.01 dBm)                                                   |

|      |                  |    |                               |
|------|------------------|----|-------------------------------|
| 0x9D | Antenna 14 Power | RW | short: read power (0.01 dBm)  |
|      |                  |    | short: write power (0.01 dBm) |
| 0x9E | Antenna 15 Power | RW | short: read power (0.01 dBm)  |
|      |                  |    | short: write power (0.01 dBm) |
| 0x9F | Antenna 16 Power | RW | short: read power (0.01 dBm)  |
|      |                  |    | short: write power (0.01 dBm) |
| 0xF0 | Preset Key 1     | WO | Preset key, UHF 4 bytes,      |
| 0xF1 | Preset Key 2     | WO | Preset key                    |
| 0xF2 | Preset Key 3     | WO | Preset key                    |
| 0xF3 | Preset Key 4     | WO | Preset key                    |
| 0xF4 | Preset Key 5     | WO | Preset key                    |
| 0xF5 | Preset Key 6     | WO | Preset key                    |
| 0xF6 | Preset Key 7     | WO | Preset key                    |
| 0xF7 | Preset Key 8     | WO | Preset key                    |
| 0xF8 | Preset Key 9     | WO | Preset key                    |
| 0xF9 | Preset Key 10    | WO | Preset key                    |
| 0xFA | Preset Key 11    | WO | Preset key                    |
| 0xFB | Preset Key 12    | WO | Preset key                    |
| 0xFC | Preset Key 13    | WO | Preset key                    |
| 0xFD | Preset Key 14    | WO | Preset key                    |
| 0xFE | Preset Key 15    | WO | Preset key                    |
| 0xFF | Preset Key 16    | WO | Preset key                    |

## 1.3 UHF Parameters

| Parameter ID | Parameter Name           | R/W | Value Description                                             |
|--------------|--------------------------|-----|---------------------------------------------------------------|
| 0x0F         | UHF Properties           | RO  | bitmap[256]: supported regions bitmap                         |
|              |                          |     | bitmap[256]: supported modes bitmap                           |
| 0x10         | UHF Frequency Parameters | RW  | byte: region                                                  |
|              |                          |     | byte: frequency hopping mode, 0=manual, 1=automatic, 2=random |
|              |                          |     | array: manual frequency hop table                             |



|      |                              |    |                                                                                  |
|------|------------------------------|----|----------------------------------------------------------------------------------|
|      |                              |    | short: hop interval (ms)                                                         |
|      |                              |    | int: start frequency (kHz)                                                       |
|      |                              |    | short: frequency step (kHz)                                                      |
|      |                              |    | byte: number of channels                                                         |
|      |                              |    | int: start frequency band 2 (kHz)                                                |
|      |                              |    | short: frequency step band 2 (kHz)                                               |
|      |                              |    | byte: number of channels band 2                                                  |
| 0x60 | UHF EPC Baseband Parameters  | RW | byte: access mode (0=Standard, 1=Single-Protocol, etc.)                          |
|      |                              |    | byte: session                                                                    |
|      |                              |    | byte: Target                                                                     |
|      |                              |    | byte: initial Q value (slot count = $2^Q$ )                                      |
|      |                              |    | byte: minimum Q value                                                            |
|      |                              |    | byte: maximum Q value                                                            |
| 0x61 | UHF EPC Algorithm Parameters | RW | byte: algorithm type                                                             |
|      |                              |    | byte: algorithm parameter 1                                                      |
|      |                              |    | short: algorithm parameter 2                                                     |
|      |                              |    | int: algorithm parameter 3                                                       |
| 0x80 | Tag Select Rule 1            | RW | byte: logic relation, 0=AND, 1=OR                                                |
|      |                              |    | short: memory bank to match, ISO/IEC 18000-6C: 0=Reserved, 1=EPC, 2=TID, 3=User. |
|      |                              |    | short: start bit address for matching                                            |
|      |                              |    | short: bit length to match                                                       |
|      |                              |    | array: data to match                                                             |
| 0x81 | Tag Select Rule 2            | RW |                                                                                  |
|      |                              |    |                                                                                  |
|      |                              |    |                                                                                  |
|      |                              |    |                                                                                  |
|      |                              |    |                                                                                  |
| 0x82 | Tag Select Rule 3            | RW |                                                                                  |
|      |                              |    |                                                                                  |
|      |                              |    |                                                                                  |
|      |                              |    |                                                                                  |

|      |                   |    |  |
|------|-------------------|----|--|
|      |                   |    |  |
| 0x83 | Tag Select Rule 4 | RW |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |
| 0x84 | Tag Select Rule 5 | RW |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |
| 0x85 | Tag Select Rule 6 | RW |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |
| 0x86 | Tag Select Rule 7 | RW |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |
| 0x87 | Tag Select Rule 8 | RW |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |
|      |                   |    |  |

## 1.4 Application Parameters

| Parameter ID | Parameter Name            | R/W | Value Description                                                                   |
|--------------|---------------------------|-----|-------------------------------------------------------------------------------------|
| 0x01         | Access Control Parameters | RW  | byte: mode, 0=disable access control, 1=enable, 2=enable with auto-read on power-on |
|              |                           |     | int: lock drive duration (ms)                                                       |

|      |                             |    |                                                                       |
|------|-----------------------------|----|-----------------------------------------------------------------------|
|      |                             |    | int: card swipe debounce interval (ms)                                |
|      |                             |    | int: door magnetic timeout (ms), door open longer than this = timeout |
| 0x02 | Auxiliary Output Parameters | RW | byte: trigger condition, 0=none, 1=aux relay input                    |
|      |                             |    | byte: aux relay action, 0=close, 1=open, 2=normally open              |
|      |                             |    | int: drive time (ms)                                                  |
| 0x08 | Door Relay GPIO             | RW | byte: door relay GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..         |
|      |                             |    | byte: type, 0=normally closed, 1=normally open                        |
| 0x09 | Door Sensor GPIO            | RW | byte: door sensor GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..        |
|      |                             |    | byte: type, 0=normally closed, 1=normally open                        |
| 0x0A | Door Open Button GPIO       | RW | byte: door open button GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..   |
|      |                             |    | byte: type, 0=normally closed, 1=normally open                        |
| 0x0B | Interlock Input GPIO        | RW | byte: interlock input GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..    |

|      |                               |    |                                                                                     |
|------|-------------------------------|----|-------------------------------------------------------------------------------------|
|      |                               |    | byte: type, 0=normally closed, 1=normally open                                      |
| 0x0C | Auxiliary Input GPIO          | RW | byte: auxiliary input GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..                  |
|      |                               |    | byte: type, 0=normally closed, 1=normally open                                      |
| 0x0D | Auxiliary Output GPIO         | RW | byte: auxiliary output GPIO index, 0=none, 1=GPIO1, 2=GPIO2, ... ..                 |
|      |                               |    | byte: type, 0=normally closed, 1=normally open                                      |
| 0x10 | MQTT Configuration            | RW | byte: mode, 0=disable MQTT, 1=enable MQTT, 2=enable MQTT with auto-read on power-on |
|      |                               |    | string: broker address                                                              |
|      |                               |    | short: broker port                                                                  |
|      |                               |    | byte: mode, 0=unencrypted, 1=SSL one-way auth                                       |
|      |                               |    | string: username                                                                    |
|      |                               |    | string: password                                                                    |
|      |                               |    | string: upstream topic                                                              |
|      |                               |    | byte: upstream QoS                                                                  |
|      |                               |    | string: downstream topic                                                            |
|      |                               |    | byte: downstream QoS                                                                |
| 0x11 | MQTT Keepalive Configuration  |    | bool: enable                                                                        |
|      |                               |    | int: keepalive interval                                                             |
| 0x12 | MQTT Tag Upload Configuration |    | bool: enable                                                                        |
|      |                               |    | int: tag upload interval                                                            |

|  |  |  |                                                                                                                          |
|--|--|--|--------------------------------------------------------------------------------------------------------------------------|
|  |  |  | int: max tags per upload                                                                                                 |
|  |  |  | int: additional data mask, bit0: antenna, bit3: TID, bit5: RSSI, bit6: frequency, bit7: phase, bit29: SN. bit30: AssetID |

## Appendix 2: Region Table

The list of supported regions is determined at device manufacture time.

| ID | Description                       | Channel Count |
|----|-----------------------------------|---------------|
| 0  | CMIIT 920~925MHz                  | 16            |
| 1  | FCC 902~928MHz                    | 50            |
| 2  | ETSI 866~867MHz                   | 4             |
| 3  | ETSI_HIGH 916~918MHz              | 3             |
| 4  | ETSI_HYBRID 866~867MHz 916~918MHz | 7             |
| 5  | KOR 917~921MHz                    | 6             |
| 6  | JPN_NO_LBT 916~921MHz             | 4             |
| 7  | JPN_LBT 916~921MHz                | 6             |
| 8  | JPN_NO_LBT_19 916~924MHz          | 19            |
| 9  | TWN 920~927MHz                    | 14            |
| 10 | MYS 919~923MHz                    | 8             |
| 11 | RUS_LOW 866~867MHz                | 5             |
| 12 | RUS_HIGH 916~920MHz               | 4             |
| 13 | AUS 920~926MHz                    | 10            |
| 14 | IND 865~867MHz                    | 4             |
| 15 | BRA 902~907.5MHz 915~928MHz       | 35            |
| 16 | VIE 918~923MHz                    | 6             |
| 17 | ISR 915~917MHz                    | 1             |

| ID  | Description       | Channel Count |
|-----|-------------------|---------------|
| 18  | ZAF 916~920MHz    | 17            |
| 19  | SGP 920~925MHz    | 10            |
| 20  | THA 920~925MHz    | 10            |
| 240 | TEST1 800~1000MHz | 197           |
| 241 | TEST2 840~960MHz  | 117           |
| 242 | TEST3 860~960MHz  | 11            |
| 243 | TEST4 860~930MHz  | 8             |

## Appendix 3: Device Status Codes

Device status code

| Value | Description                          |
|-------|--------------------------------------|
| 0x00  | Success                              |
| 0x01  | Invalid command type                 |
| 0x02  | Invalid command length               |
| 0x03  | Invalid command parameter            |
| 0x04  | Command execution error              |
| 0x05  | Command no response                  |
| 0x11  | GPIO index out of range              |
| 0x12  | Invalid GPIO mode                    |
| 0x13  | Other GPIO error                     |
| 0x61  | Invalid parameter type               |
| 0x62  | Invalid parameter value              |
| 0x63  | Invalid parameter length             |
| 0x64  | Failed to get parameter              |
| 0x65  | Failed to set parameter              |
| 0x66  | Failed to save parameter             |
| 0x67  | Other parameter error                |
| 0x80  | Tag returned other error             |
| 0x81  | Tag returned not supported           |
| 0x82  | Tag returned insufficient privileges |

|      |                                        |
|------|----------------------------------------|
| 0x83 | Tag returned memory overrun            |
| 0x84 | Tag returned memory locked             |
| 0x85 | Tag returned cryptographic suite error |
| 0x86 | Tag returned command not encapsulated  |
| 0x87 | Tag returned response buffer overflow  |
| 0x88 | Tag returned security timeout          |
| 0x8B | Tag returned insufficient power        |
| 0x8E | Tag lost                               |
| 0x8F | Non-specific error                     |
| 0x90 | Invalid RFID region                    |
| 0x91 | Invalid RFID value                     |
| 0x9A | RFID execution error                   |
| 0x9B | RFID no acknowledgment                 |
| 0x9E | RFID internal error                    |
| 0x9F | Other RFID error                       |
| 0xFB | Device busy                            |
| 0xFE | Internal error                         |
| 0xFF | Invalid connection                     |